

Pratyush Agrawal

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Education

Manipal Institute of Technology,
B.Tech in Information Technology – Karnataka, India

2024 – 2028
CGPA: 7.86 / 10

Coursework

Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Object-Oriented Programming, Discrete Mathematics, Computer Organization

Projects

KV Cache Quantization | *Python, PyTorch, Hugging Face Transformers, LLM Optimization* [GitHub](#) | [Blog](#)

- Reduced GPT-2 KV cache footprint by **68.5%** with only a **0.71% increase in perplexity** using a KIVI-inspired INT8 quantization pipeline.
- Architected a custom inference engine in PyTorch, bypassing `model.generate()` to implement manual cache management and token-by-token generation for granular control over tensor states.
- Engineered flexible quantization schemes supporting symmetric/asymmetric modes across whole-tensor, per-channel, and per-token granularities to evaluate reconstruction error on real GPT-2 tensors.
- Conducted comprehensive benchmarking of FP32, FP16, and INT8 formats, analyzing the trade-offs between inference latency, memory consumption, and teacher-forced perplexity.

DataTalk AI | *React.js, Node.js, Azure OpenAI, Cosmos DB, REST API* [GitHub](#) | [Demo](#)

- Developed a multimodal retail AI that ranked in the **Top 50 of 300+ teams** at the Microsoft AI Unlocked Hackathon by automating bill generation from unstructured speech and imagery.
- Designed a high-concurrency Node.js orchestration layer to sequence Azure Speech-to-Text and Computer Vision into a unified NLP context window, persisting structured JSON data to Cosmos DB.
- Refined data extraction accuracy through iterative prompt engineering with Azure OpenAI, successfully parsing itemized quantities and prices from noisy, colloquial kirana store audio.
- Built a real-time React.js dashboard for live-streaming mic and webcam feeds, integrating WhatsApp Deep Linking for automated, instant digital bill delivery to customers.

Achievements

Microsoft AI Unlocked Hackathon 2025

- Ranked in the top 50 out of 300+ teams nationwide with DataTalk AI

Skills

Languages: Python (Proficient), Java(Proficient), JavaScript, C, SQL

AI/ML: PyTorch, Hugging Face Transformers, LLM Inference Optimization, Quantization, Prompt Engineering, Multimodal Fusion

Frameworks & Libraries: React.js, Next.js, Node.js, Express.js, Django, Django REST Framework, Tailwind CSS

Databases: PostgreSQL, Azure Cosmos DB, NoSQL

Cloud & DevOps: Azure, Git, GitHub, Vercel, Render, REST APIs

Extra Curricular

IEEE Student Branch, Manipal Institute of Technology 2024 – Present

- Managing Committee Member and WebDev Domain POC for TechWeek-IEEE SBM's flagship technical fest
- Co-organized DevRush hackathon with 18 participating teams